

A general framework for using two-stage IPD-MA to predict individualized treatment effects

Marta Mainetti

University of Bern

ISCB conference 2026

u^b

u^b
UNIVERSITÄT
BERN

Background: Individualized Treatment Effect

In meta-analyses, the aim is to estimate an average treatment effect using aggregated data from randomized trials (RCTs)

Individualized Treatment Effect (ITE)

- however, the effect of a treatment can **vary across patients**
- we define *Individualized Treatment Effect* the **patient-level benefit** gained from being treated
- how can we make predictions about it?



- data from single RCTs are not suitable
- we need multiple trials to investigate effect modification and make predictions

Individual Participant Data Meta-Analysis (IPD-MA)

- **patient-level data** collected from multiple randomized controlled trials with similar research question
- *meta-analysis*, i.e. only two interventions are compared
- can be conducted with a **one-stage** or a **two-stage approach**

One-stage approach

- centralizes data and conducts a **single analysis**
- infeasibility with data-sharing constraints

Two-stage approach

1. analyses are performed **separately** for each study
2. study-specific results are pooled and synthesized

Aims of the study

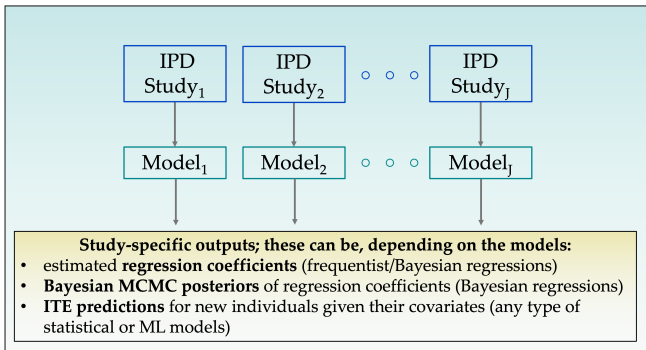
1. present a **general** two-stage IPD meta-analysis **framework** for predicting **ITE**
 - **Within-trial flexibility** - any predictive model for ITE can be employed in first stage
 - **Between-trial adaptability** - different models can be used across studies
2. evaluate performances with a **simulation study** for a continuous outcome

Framework pipeline: a two-stage approach (I)

First stage: fit study-specific models.

This can be done with **any type of model**:

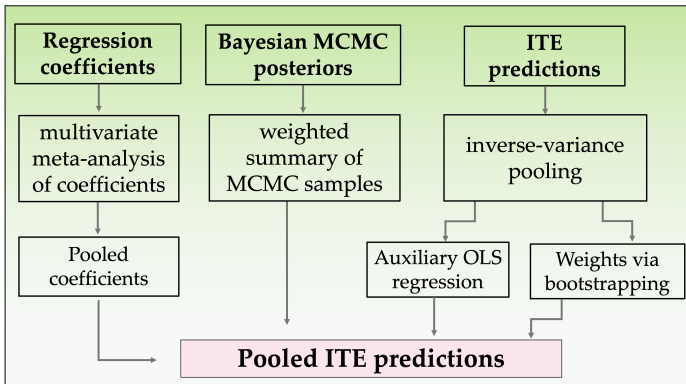
- Frequentist regression models (OLS, LASSO, Ridge)
- Machine learning (XGBoost, Causal Forest)
- Bayesian regression models
- Meta-learners (R- and X-)



Framework pipeline: a two-stage approach (II)

Second stage: pooling of study-specific results

- We used different pooling methods to yield ITE predictions
- Applicable second stage methods **depends on the choice of first-stage models**



Data generation mechanisms (DGMs), varying:

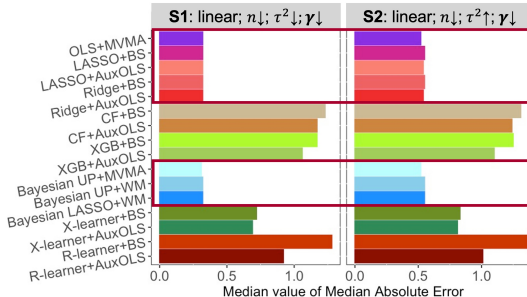
- study-specific sample size (n)
- linear or complex treatment-covariates interactions
- heterogeneity of the main treatment effect (τ^2)
- magnitude of the linear effect modification (γ)
- ratio for treatment allocation (1:1, 2:1)

Performance was measured with Median Absolute Error (MAE), Mean Bias and R^2

Simulation results: Models comparison (I)

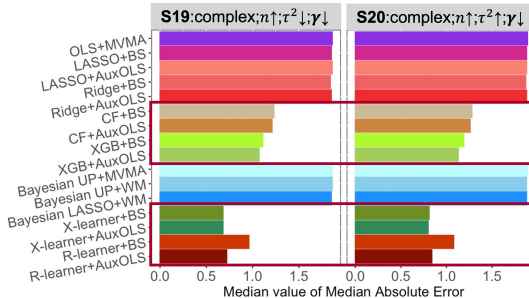
Simulations showed that the performance of *framework specifications* depended on the underlying DGM

- In scenarios with linear interactions and small samples, **regression-based models** in the first stage consistently outperformed machine learning



Simulation results: Models comparison (II)

- With complex interactions and large samples, **machine learning** approaches performed better than linear regressions
- In particular, **weighting by OLS inverse-variance** in the second stage performed better than bootstrapping



Conclusions

- We developed a framework that leverages two-stage IPD-MA to predict individualized treatment effect and tested it in a simulation study
- The framework is flexible and adaptable
- Simulations showed how performance depended on the way data were generated, for example:
 - linear models work well with small samples
 - with larger samples and complex effect modifications, machine learning lead to more accurate predictions
 - meta-learners (R- and X-) with XGBoost performed even better
- However, data generation mechanisms are not known in real world applications. Specification of the framework and model selection should be based on proper validation

Thank you for your attention!

Publication

A general framework for using two-stage meta-analysis with individual participant data to predict individualized treatment effects

Marta Mainetti¹ [Email](#)

Konstantina Chalkou¹

Frederik Luca Philipona¹

Guido Schwarzer²

Georgia Salanti¹

Orestis Efthimiou¹



Contact

marta.mainetti@unibe.ch

Funding

